POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2021
Programme: BE | Full Marks: 100
Course: Digital Signal Analysis and Processing | Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define energy and power signal with examples. Check whether the given signal is energy or power signal x(t)=Asin(t). 8

   b) Prove that an LTI system is causal if the impulse response of the system is a causal signal. 7

2. a) Find the convolution between two signals $x[n] = a^n, for\ 0 \leq n \leq 6$ and $h[n] = 1, for\ 0 \leq n \leq 4$ 8

   b) Define Z-transform. Find Z-transform of signal $x[n]= (\frac{1}{2})^n u[n]+(1+j)^n u[-n-1]$ and indicate ROC graphically. 7

3. a) Why DFT is needed and how the DFT solve the problem associated with DTFT. Use DIF-FFT algorithm to compute 8 point DFT of x[n]={2,1,1,1}. Discuss the result. 8

   b) Show that the multiplication of two DFT sequences results in circular convolution. 7

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4. a) Realize the transfer function using cascade realization 7

$$X(Z)=\frac{2(z+2)}{z(z-0.1)(z+0.5)(z-0.4)}$$

   b) Draw the lattice structure of the given transfer function 8

$$H(Z)= 1+\frac{3}{8}z^{-1}+\frac{5}{4}z^{-2}+\frac{3}{2}z^{-3}$$

5. a) Design a FIR system to meet the following specifications 8

   Pass band edge frequency=2KHz

   Stop band edge frequency=5KHz

   Stop band attenuation=42dB

   Sampling frequency=20KHz

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7 b) Derive Impulse Invariant Method for IIR filter design. Also illustrate the mapping from the s-plane to the z-plane while using IIM.

6. a) Design a second order discrete-time Butterworth filter with cut-off frequency of 1 KHz and sampling frequency of 10^4 samples/sec by using bilinear transformations. Also plot the poles of the filter.

6.3 b) Design a linear FIR filter using Kaiser window to meet the following specifications:

$$0.99 \leq |H(e^{j\omega})| \leq 1.01, for 0 \leq |\omega| \leq 0.19\pi$$
$$\leq |H(e^{j\omega})| \leq 0.01, for 0.21\pi \leq |\omega| \leq \pi$$

7. Write short notes on: (**Any two**)
   a) Region of Convergence 3.2
   b) Causal and Non-Causal system 2
   c) Gibbs Phenomena 6.1

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